

DM-ASR: Diarization-aware Multi-speaker ASR with Large Language Models

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Abstract

Multi-speaker automatic speech recognition (ASR) aims to transcribe conversational speech involving multiple speakers, requiring the model to capture not only what was said, but also who said it and sometimes when it was spoken. Recent Speech-LLM approaches have shown the potential of unified modeling for this task, but jointly learning speaker attribution, temporal structure, and lexical recognition remains difficult and data-intensive. At the current stage, leveraging reliable speaker diarization as an explicit structural prior provides a practical and efficient way to simplify this task. To effectively exploit such priors, we propose DM-ASR, a **Diarization-aware Multi-speaker ASR** framework that reformulates the task as a multi-turn dialogue generation process. Given an audio chunk and diarization results from a dedicated diarization system, DM-ASR decomposes transcription into a sequence of speaker- and time-conditioned queries, each corresponding to one speaker in one time segment. This formulation converts multi-speaker recognition into a series of structured sub-tasks, explicitly decoupling speaker-temporal structure (who and when) from linguistic content (what), and enabling effective integration of diarization cues with the reasoning capability of large language models. We further introduce an optional word-level timestamp prediction mechanism that interleaves word and timestamp tokens, yielding richer structured outputs and better transcription quality. Our analysis shows that diarization systems provide more reliable speaker identities and segment-level boundaries, while LLMs excel at modeling linguistic content and long-range dependencies, demonstrating their complementary strengths. Experiments on both Mandarin and English benchmarks show that the proposed approach achieves strong performance with relatively small models and training data, while remaining competitive with or outperforming existing unified approaches.

CCS Concepts

• **Computing methodologies** → **Speech recognition**.

Keywords

Multi-speaker ASR, Speech-LLM, Speaker Diarization

1 Introduction

Multi-speaker Automatic Speech Recognition (ASR) aims to transcribe conversational speech involving multiple participants. In practical scenarios such as meetings, interviews, lectures, and call-center conversations, users care not only about *what* was said, but also *who* said it and, in many cases, *when* it was spoken [14, 26, 56]. Compared with conventional single-speaker ASR, this setting is substantially more challenging, since lexical recognition, speaker attribution, and sometimes temporal localization must be addressed jointly under rapid turn-taking, speaker overlap, and long-range conversational context.

Existing systems for multi-speaker ASR can be broadly grouped into four categories. The first category is *cascaded diarization-ASR pipelines* [1, 11, 44, 57], where speaker diarization first estimates *who spoke when*, and ASR is then applied to diarized segments, followed by alignment or stitching to form speaker-attributed transcripts. While such pipelines are modular and practical, they are inherently vulnerable to error propagation, speaker-text mismatch, and boundary inconsistency, especially in overlapping and long-form conversations [12, 38]. The second is *jointly trained speaker-attributed ASR models* [24, 25, 29, 39, 46, 58, 59], which remain within a conventional ASR framework but model speaker attribution and recognition more tightly, often in parallel, to reduce the mismatch introduced by cascaded designs. The third category is *LLM-assisted hybrid systems* [4, 13, 40, 48, 51], which preserve the hybrid pipeline structure while introducing an LLM to globally refine speaker attribution and transcript consistency. The fourth category is *unified end-to-end Speech-LLM systems* [21, 35, 42, 45, 54, 55], where multi-speaker speech understanding is directly formulated as structured generation within a single model.

Although recent progress has been substantial, two issues remain insufficiently explored. First, many existing methods mainly target *who said what*, while explicit modeling of *when* is still weakly represented in a large portion of the literature [21, 42]. However, explicit temporal grounding is essential for realistic meeting transcription, downstream retrieval, and standard diarization-oriented evaluation. Second, while unified end-to-end Speech-LLM systems are a promising long-term direction for multi-speaker ASR, explicit diarization information remains highly valuable, especially when model size or training data is limited. In these settings, requiring a

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Table 1: Comparison of recent works on Multi-speaker ASR by explicit text outputs.

Recent Work on Multi-speaker ASR	Use LLM	Speaker	Segment-level Timestamp	Word-level Timestamp
Sortformer [39]; Meta-CAT [46]; DNCASR [59]	✗	✓	✗	✗
MT-LLM [36]; DiarizationLM [48] (post-processing only)	✓	✗	✗	✗
SpeakerLM [54]; JEDIS-LLM [45]	✓	✓	✗	✗
Tagspeech [21]; VibeVoice-ASR [42]; Moss transcribe diarize [55]	✓	✓	✓	✗
DM-ASR (Ours)	✓	✓	✓	✓

model to learn speaker attribution, temporal grounding, and lexical recognition entirely from raw supervision can be unnecessarily difficult. Therefore, at the current stage, leveraging reliable speaker diarization as an explicit structural prior provides a practical and efficient way to simplify this multi-speaker ASR task.

Motivated by these observations, we propose **DM-ASR**, a Diarization-aware Multi-speaker ASR framework. DM-ASR can be regarded as a form of diarization-ASR pipeline, but unlike traditional cascaded systems, it does not use diarization to split audio into speaker-wise utterances for a single-speaker ASR backend. Instead, it uses diarization as an explicit structural prior while allowing an LLM-based backend to recognize mixed multi-speaker speech directly. This avoids the difficulty of speaker-wise segmentation under overlap and preserves longer conversational context during decoding. With this design, DM-ASR enables smaller models to achieve strong performance without relying solely on model size and data scaling. It further supports word-level timestamp prediction, yielding richer structured outputs and better transcription quality. To better understand the interaction between external diarization cues and model inference, we further design multiple evaluation settings covering different combinations of diarization-provided and LLM-predicted speaker and times information. Results show that the model can effectively exploit reliable diarization cues and gradually learn to correct imperfect ones as model size and training data increase. Our contributions are as follows:

First, we propose DM-ASR, a diarization-aware multi-speaker ASR framework that reformulates multi-speaker transcription as a multi-turn dialogue generation process. By explicitly incorporating diarization cues into decoding, it achieves strong performance with a smaller model and less training data.

Second, we enable word-level timestamp prediction and show that temporally grounded generation improves not only output structure but also text accuracy. We further introduce multiple evaluation settings to analyze when the model should rely on diarization priors and when it can correct them.

Third, we validate the proposed method on both Mandarin and English datasets, demonstrating its effectiveness and generalizability across languages and conversational conditions.

2 Related Work

2.1 Speaker Diarization Systems

Speaker diarization aims to determine *who spoke when* in multi-speaker speech and serves as a fundamental component of conversational speech understanding [23, 41]. Early studies on speaker

diarization focus on modularized pipelines [27, 30, 47], which first segment the audio and then cluster the segments based on speaker embedding similarity. Since these methods typically assume that each segment contains only one active speaker, their performance degrades substantially in overlapping speech. To address this limitation, End-to-End Neural Diarization (EEND) formulates diarization as a multi-label prediction problem, enabling more robust modeling of overlapping speech [15, 19, 20]. Building on this line of work, Pyannote [2, 3] has shown strong performance across multiple benchmarks. DiariZen [17, 18] further improves upon Pyannote by incorporating pretrained WavLM representations [5], thereby benefiting from richer acoustic information. In parallel, Target-Speaker Voice Activity Detection (TSVAD) combines modular and neural designs and has achieved strong performance in practical diarization systems [8, 34, 49, 50]. More recently, Sequence-to-Sequence Neural Diarization (S2SND) [6, 7] has advanced the online diarization setting through a sequence generation formulation.

2.2 Multi-speaker ASR: Cascaded and Joint Modeling

A common solution to multi-speaker ASR is the cascaded pipeline [1, 11, 44, 57], in which speaker diarization is first performed to estimate speaker turns and boundaries, and ASR is then applied to the resulting segments. While effective and easy to deploy, such systems suffer from error propagation, speaker-text mismatch, and limited ability to exploit the mutual dependence between speaker structure and lexical content [12, 38]. These issues become more severe in overlapping, noisy, or long-form conversations.

To mitigate these limitations, a line of work has moved toward jointly trained speaker-attributed ASR, where speaker modeling and recognition are optimized in a more unified manner. Early efforts explored joint speech recognition and speaker diarization through sequence transduction and related integrated formulations [23]. This direction was later extended to end-to-end and transcribe-to-diarize style modeling, further tighten the coupling between speaker modeling and recognition [24, 25, 29, 39, 46, 59]. As summarized in Table 1, these methods such as Sortformer [39], Meta-CAT [46], and DNCASR [59] mainly focus on recovering *who said what*, rather than the more complete objective of *who said what when*.

2.3 LLM-based Multi-speaker ASR

Recent advances in large language models (LLMs) have opened a new direction for multi-speaker ASR. One line of work uses

LLMs as post-processing or reconciliation modules on top of conventional diarization-ASR pipelines. A representative example is DiarizationLM[48], which refines the outputs of independent ASR and diarization systems to improve transcript consistency and speaker attribution. As shown in Table 1, such methods may benefit from LLM-based reasoning, but they still do not explicitly generate speaker-aware timestamps, and remain constrained by the quality of upstream diarization and ASR outputs.

Another line of work formulates multi-speaker ASR as structured generation within a unified end-to-end Speech-LLM. Representative systems include SpeakerLM, JEDIS-LLM, MOSS Transcribe Diarize, TagSpeech, and VIBEVOICE-ASR [21, 42, 45, 54, 55]. Compared with earlier approaches, these methods better exploit long-range conversational context and enable more flexible structured decoding. However, Table 1 also highlights a clear distinction within this family. Some methods, such as SpeakerLM and JEDIS-LLM, mainly target speaker-attributed transcription, i.e., *who said what*, without explicit timestamp prediction [45, 54]. In contrast, more recent systems such as TagSpeech, MOSS Transcribe Diarize, and VIBEVOICE-ASR move toward the richer objective of *who said what when* by explicitly generating segment-level timestamps [21, 42, 55].

However, these methods do not explore word-level timestamp prediction, a standard capability in conventional ASR systems such as Whisper [43]. In addition, although unified Speech-LLMs have shown strong potential, they typically rely on large models ($\geq 7B$) and broad training data, while the role of explicit diarization cues remains underexplored. Our work is closely related to the latter direction, as summarized in Table 1, but differs in two aspects. First, DM-ASR is a semi-end-to-end multi-speaker ASR framework that decouples diarization and LLM-based recognition, while using diarization outputs as explicit structural prompts for multi-turn generation. Second, it explicitly investigates diarization-aware generation under limited model sizes and data scales. Although [31] is also diarization-aware, our method directly uses raw diarization outputs and discrete prompt tokens without speaker embeddings, and further supports timestamp prediction.

3 Method

3.1 Overview

Our key idea is to reformulate multi-speaker ASR as a *multi-turn dialogue generation* problem guided by diarization cues. Instead of directly predicting a single mixed transcript from the input conversational speech, we decompose the task into a sequence of speaker- and time-conditioned transcription turns. In each turn, the model is asked to transcribe the speech content of a target speaker within a specified time segment, and the corresponding transcription is generated autoregressively by the LLM.

As shown in Figure 1, DM-ASR consists of four main components: (1) a **speech encoder** that extracts frame-level acoustic representations from multi-speaker audio, (2) a **projector** that maps speech features into the embedding space of the LLM, (3) a **large language model decoder** that performs dialogue-style structured generation, and (4) a **special-token discretization mechanism** that converts diarization outputs into speaker and timestamp tokens understandable to the LLM.

3.2 Special Tokens and Diarization-Aware Prompting

3.2.1 Special token design. DM-ASR introduces three types of special tokens to encode speaker and temporal information.

- **Speaker tokens:** a set of speaker index tokens $\{<|spk_idx_0|>, <|spk_idx_1|>, \dots\}$. Since absolute speaker identities are not semantically meaningful across utterances, we remap the original diarization labels to local speaker indices based on their order of appearance within each audio chunk. This chunk-level relabeling reduces vocabulary sparsity and simplifies the generation target. Meanwhile, we preserve the mapping from local speaker indices to absolute speaker identities for each chunk, so that the predictions can be converted back to the original speaker labels during inference.
- **Timestamp tokens:** continuous timestamps are discretized into integer indices using a fixed temporal resolution Δt :

$$u = \text{round}(t/\Delta t) \quad (1)$$

where t is the time in seconds and u is the discretized timestamp index. In our implementation, $\Delta t = 0.1$ s. The corresponding timestamp vocabulary includes tokens such as $<|time_idx_0|>$, $<|time_idx_1|>$, and so on.

- **Control tokens:** task-related control tokens, including audio delimiters, speaker delimiters, time delimiters and word-level timestamp trigger, such as $<|start_of_audio|>$, $<|start_of_spk|>$, $<|start_of_time|>$, $<|with_timestamps|>$.

All special tokens are added to the LLM vocabulary, and their embeddings are learned during training.

3.2.2 Diarization-aware prompt construction. Assume the speaker diarization system produces a set of segment labels:

$$\mathcal{D} = \{(s_k, t_k^{\text{st}}, t_k^{\text{ed}})\}_{k=1}^K \quad (2)$$

where s_k is the speaker label of the k -th segment, and $t_k^{\text{st}}, t_k^{\text{ed}}$ are its start and end times. After local speaker mapping and timestamp discretization, each segment is converted into a structured condition $(\hat{s}_k, \hat{u}_k^{\text{st}}, \hat{u}_k^{\text{ed}})$, where $\hat{s}_k$ is the local speaker index and $\hat{u}_k^{\text{st}}, \hat{u}_k^{\text{ed}}$ are the discretized start and end time indices:

$$\hat{u}_k^{\text{st}} = \text{round}(t_k^{\text{st}}/\Delta t), \quad \hat{u}_k^{\text{ed}} = \text{round}(t_k^{\text{ed}}/\Delta t) \quad (3)$$

For each segment, we construct a natural-language prompt of the following form:

Please transcribe the speech content of speaker $<|start_of_spk|><|spk_idx_s_k|><|end_of_spk|>$ within the time segment $<|start_of_time|><|time_idx_u_k^{\text{st}}|><|time_idx_u_k^{\text{ed}}|><|end_of_time|>$ into text.

This prompt explicitly provides speaker and temporal conditions, enabling unified speaker- and time-aware transcription.

3.2.3 Optional word-level timestamps. In addition to segment-level transcription, DM-ASR supports an optional fine-grained timestamp generation mode. When the control token $<|with_timestamps|>$ is appended to the prompt, the model generates the transcription in an interleaved format consisting of words and timestamp tokens:

$$T_k = [\tau_1, w_1, \tau_2, w_2, \dots, \tau_n, w_n, \tau_{n+1}] \quad (4)$$